



✅ **task breakdown** for building **e-commerce site backend** over **3 sprints in 7 days** with **Google Firebase** for data persistence and **GitHub** for project management and tracking.

Sprint 1: Backend Foundation & Database Setup (Days 1-2)

Goal: Set up the backend, create the Firebase database, and build the basic API structure.

Ticket #	Assigned To	Task Description	Deadline
1	Dev 1	Set up the GitHub repository with proper structure and invite all team members.	Day 1
2	Dev 2	Create a basic Express.js server with MVC structure. Define folder architecture.	Day 1
3	Dev 3	Configure Google Firebase and set up database rules and structure.	Day 1
4	Dev 4	Design the User model schema in Firebase (username, password, email, etc.).	Day 1
5	Dev 5	Design the Product model schema in Firebase (name, description, price, etc.).	Day 1
6	Dev 6	Create Firebase collections and ensure relationships (Users, Products, Transactions) are properly structured.	Day 2
7	Dev 1 & Dev 2	Build User CRUD API routes (Create, Read, Update, Delete) with Firebase integration.	Day 2
8	Dev 3 & Dev 4	Build Product CRUD API routes with Firebase integration.	Day 2

✅ **Sprint 1 Deliverables:**

- GitHub repository created with team access.
- Backend server running with basic MVC architecture.
- Firebase database set up with proper collections.
- Functional CRUD APIs for Users and Products.



⚙️ Sprint 2: Authentication, API Testing & Design Document (Days 3-4)

Goal: Add authentication, secure the backend, and create the design document.

Ticket #	Assigned To	Task Description	Deadline
9	Dev 1	Add JWT authentication for login and user registration.	Day 3
10	Dev 2	Hash passwords with bcrypt before saving to Firebase.	Day 3
11	Dev 3	Implement authorization middleware to protect certain routes.	Day 3
12	Dev 4	Test the API routes using Postman or Swagger. Verify all operations are functional.	Day 4
13	Dev 5 & Dev 6	Create the External Design Document with: team name, logo, table of contents, wireframes, and initial screenshots.	Day 4

✅ Sprint 2 Deliverables:

- Authentication (JWT + bcrypt) added.
- Authorization middleware implemented.
- API tested with Postman.
- External design document created.

🚦 Sprint 3: Finalization, Documentation & Presentation (Days 5-7)

Goal: Finalize the project, complete documentation, and prepare the presentation.

Ticket #	Assigned To	Task Description	Deadline
14	Dev 1 & Dev 2	Refactor and clean up the backend code . Improve error handling and validation.	Day 5
15	Dev 3	Write the final API documentation with setup instructions and usage examples.	Day 5



Ticket #	Assigned To	Task Description	Deadline
16	Dev 4 & Dev 5	Create the presentation video demonstrating the project, CRUD operations, and authentication flow.	Day 6
17	Dev 6	Design a PowerPoint or Google Slides deck with two slides summarizing the project.	Day 6
18	Dev 1, Dev 2 & Dev 3	Perform end-to-end testing and fix any final bugs.	Day 7
19	All Devs	Do a final code review , clean up unnecessary files, and prepare for presentation.	Day 7

✅ Sprint 3 Deliverables:

- Refined, production-ready backend code.
- API documentation finalized.
- Presentation video recorded.
- PowerPoint/Google Slides deck completed.
- End-to-end testing and bug fixes.

📊 Final Summary:

- **Sprint 1 (Days 1-2):** Backend setup, Firebase configuration, and basic CRUD API.
- **Sprint 2 (Days 3-4):** Authentication, authorization, and design document.
- **Sprint 3 (Days 5-7):** Final testing, documentation, and presentation.

✅ This plan ensures the team delivers a **fully functional, production-ready e-commerce backend** within **7 days** using **Google Firebase**, with clear ownership of tasks and efficient collaboration. Open to adjustments until final deliverable. Let me know if you need further adjustments or details! 🚀